October 15, 2021

Water pouring

$$
\begin{tikzcd}
 & & & (4,7,0) \arrow[dll] \arrow[drr] & & & \\
 & (0,7,4)^{\boxed{1}} \arrow[dl] \arrow[dr] & & & & (4,0,7) \arrow[dl] \arrow[dr] & \\
(4,3,4) \arrow[d] & & (0,1,10)^{\boxed{2}} \arrow[d] & & (0,4,7) \arrow[d] & & (1,0,10) \arrow[d] \\
(0,3,8) \arrow[d] & & (4,1,6)^{\boxed{3}} \arrow[d] & & (4,4,3) & & (1,7,3) \\
(3,0,8) & & (0,5,6)^{\boxed{4}} & & & &
\end{tikzcd}
$$

$(3,0,8)$ $\downarrow$ $(3,7,1)$ $\downarrow$ $(4,6,1)$

$(0,5,6)$ $\boxed{4}$ $\downarrow$ $(4,5,2)$ $\boxed{5}$ $\downarrow$ $\underline{\underline{(2,7,2)}}$ $\boxed{6}$

(repetation only)

6 steps

Depth-first search

Let $G=(V,E)$ be a directed graph given by adjacency lists.

Now the input doesn't contain a starting vertex!

Choose a starting vertex v and explore the graph from v using the directed edges. We simply go forward until we can discover new vertices. When we get stuck we step back and continue from the "previous" vertex. If we have discovered all vertices that can be discovered from v and there are still undiscovered parts of the graph we choose another starting vertex and continue

The algorithm terminates when we discov-ered the whole graph

Organization

Similar to the BFS but here we use a stack to store the grey vertices

stack $\longleftrightarrow$ recursive description

Attributes (vertices)

- colours : white, grey, black

undiscovered discovered finished

- parent (c.f. BFS)

- discovery time $\to$ $s[v]$ (white $\to$ grey)
  finishing time $\to$ $f[v]$ (grey $\to$ black)

  $2|V|$ timestamps : $1, 2, \ldots, 2|V|$

  $$1 \leq s[v] < f[v] \leq 2|V|$$

DFS forest :
composed of the edges $(\text{parent}[u], u)$

DepthFirstSearch(G)

```
for each u in V do
    colour[u] := white
    parent[u] := nil
    s[u] := 0
    f[v] := 0
time := 0
for each u in V do
    if colour[u] = white then DFS(u)
```

explore from u

```
DFS (u)

time := time + 1
colour[u] := grey
s[u] := time
for each v in AdjList[u] do
    if colour[v] = white then
        parent[v] := u
        DFS (v)

time := time + 1
colour [u] := black
f[u] := time
```

Cost

$O(|V|+|E|)$ because DFS(u) is called once for each u and we scan AdjList$[u]$ once in this call.

Classification of the edges

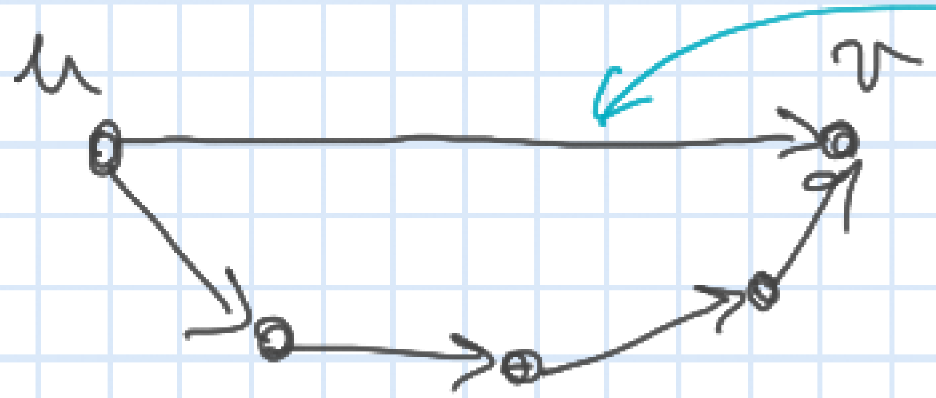
(with respect to an execution of the DFS algorithm on G)

An edge (u,v) is a

Ⓐ tree edge → an edge of the DFS tree

$s[v] = 0$ when we "discover" (u, v)

Ⓑ forward edge →

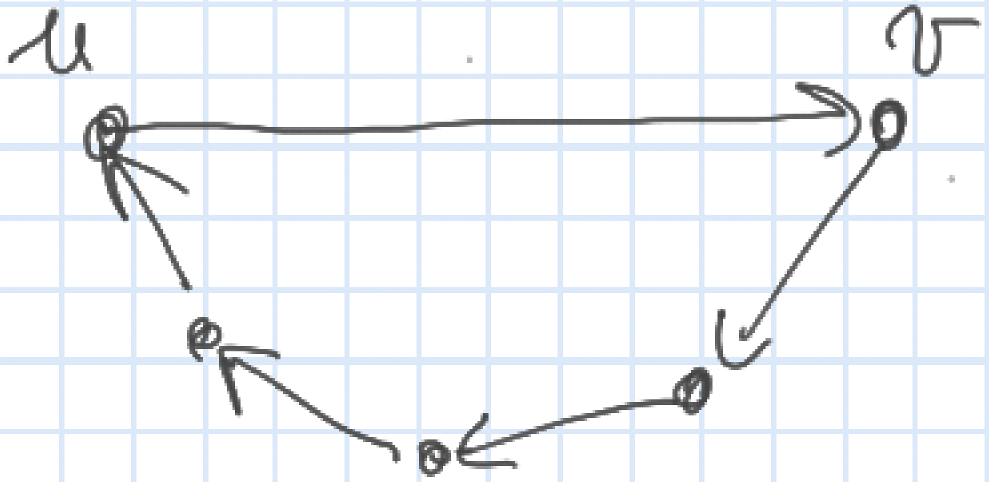


not a tree edge!

tree edges

$s[u] < s[v]$ when we "discover" (u, v)

Ⓒ back edge →



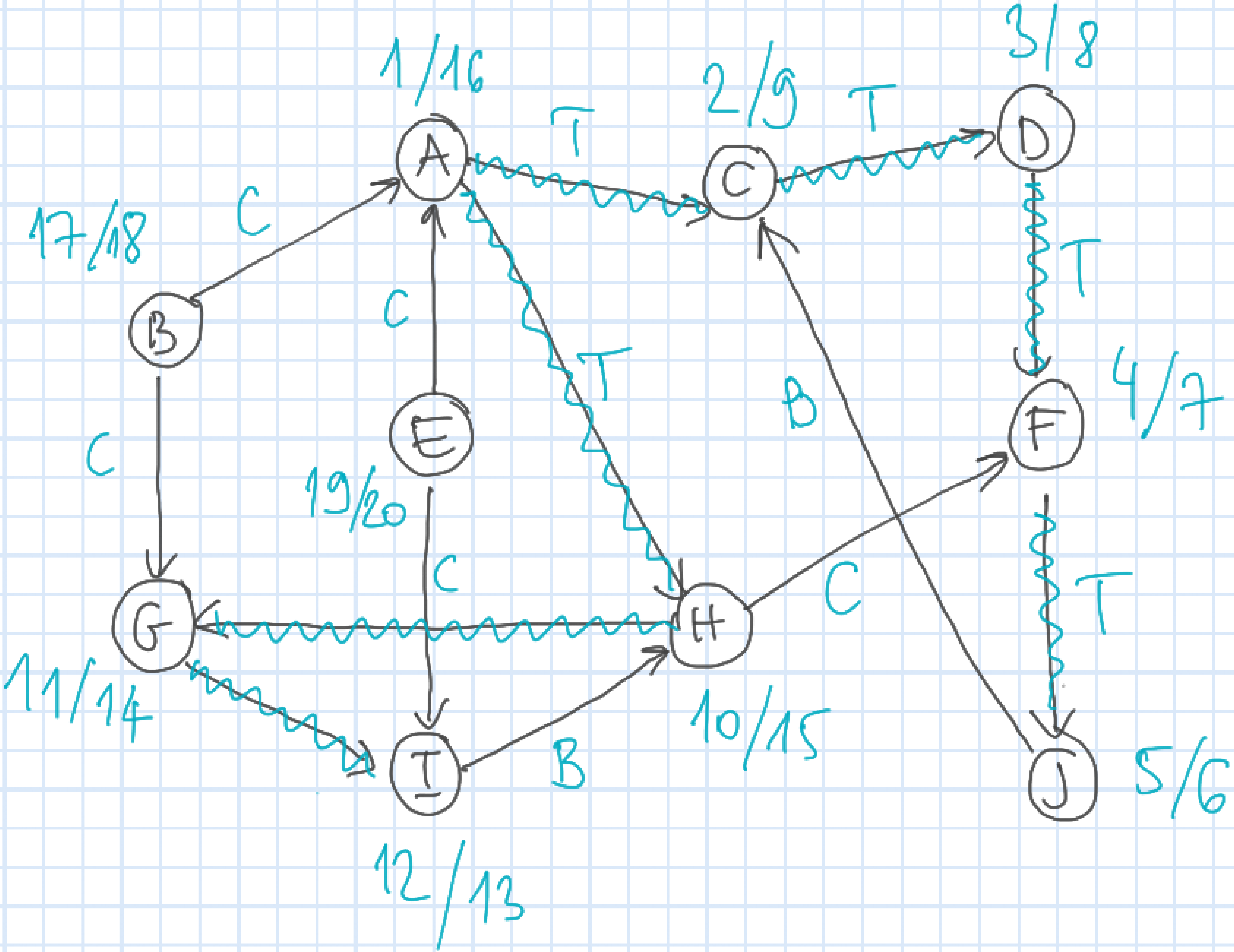
tree edges

$s[u] > s[v]$ and $f[v] = 0$ when we discover (u,v)

(D) cross edge → all others

$s[u] > s[v]$ and $f[v] > 0$ when we discover (u,v)

Sample execution of DFS



Quiz Problem!

CANVAS